BOLTPE FINANCIAL PLANNER

Functional Specification Document (FSD)

Independent Financial Planning Engine
Without Policy Planner Runtime Dependency

Document Attribute	Value
Document Version	1.2
Product	BoltPe Financial Planner
Document Type	Functional Specification Document
Primary Objective	Build BoltPe-owned financial planning logic that reproduces the existing output behavior for the same inputs and assumptions, without calling Policy Planner APIs.
Status	Proposed / Implementation Baseline
Prepared For	BoltPe Financial Planner Product & Engineering Team

Source basis: current BoltPe Financial Planner API documentation and the existing normalized response contract. Exact formula parity for opaque vendor rules must be validated through test cases before production sign-off.

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1. Executive Summary

BoltPe Financial Planner will be an independent financial-planning product in which BoltPe owns the complete calculation and recommendation engine. The system will no longer depend on the Policy Planner API for runtime calculation, scoring, insurance analysis, feasibility, recommendations, or report-generation data.

The first implementation goal is behavioral parity with the current normalized response: when the same customer input, selected goals, timelines, inflation assumptions, return assumptions, insurance inputs and risk profile are provided, the BoltPe engine should produce the same or approved-equivalent calculated values.

Key Principle	Policy Planner is removed from the runtime path. BoltPe becomes the source of truth for all financial calculations. Exact parity is guaranteed only for rules that have been explicitly derived, documented and regression-tested.

2. Purpose and Objectives

- Remove the operational dependency on Policy Planner APIs.
- Keep the existing BoltPe customer journey and normalized report structure as stable as possible.
- Reproduce the same goal-planning outputs using BoltPe-owned calculations.
- Own the formulas, assumptions, scoring logic, insurance rules, feasibility logic and recommendation rules.
- Make calculations testable, auditable, versioned and independent of PDF or email generation.
- Allow future product enhancements without waiting for a third-party vendor.

3. Scope

3.1 In Scope

- Customer and family profile capture
- Cashflow analysis
- Instant, short-term and long-term goal planning
- Inflation-based future corpus calculation
- Lump-sum requirement calculation
- Monthly saving / SIP requirement calculation
- Goal feasibility and achievable timeline analysis
- Financial score and financial status
- Insurance requirement and protection-gap analysis
- Risk-based asset allocation
- Priority roadmap and recommendations
- Structured JSON response
- BoltPe PDF generation
- Database persistence
- S3 / object storage integration and signed links
- Email delivery
- Calculation audit, assumption versioning and regression testing

3.2 Out of Scope for Initial Parity Release

- Real-time market-price recommendations
- Execution of mutual fund, insurance or securities transactions
- Automated tax filing
- Regulated personalized investment advisory beyond approved product rules
- Any undocumented proprietary vendor algorithm that cannot be inferred and validated from available inputs/outputs

4. Existing and Target Architecture

4.1 Current Runtime Flow

```
Frontend
  -> BoltPe API
  -> Input Validation
  -> Policy Planner API
  -> Vendor Calculation
  -> Vendor Response
  -> BoltPe Normalization
  -> Database
  -> PDF / S3 / Email
```

4.2 Target Runtime Flow

```
Frontend
  -> BoltPe API
  -> Input Validation
  -> BoltPe Planning Engine
       -> Profile & Cashflow Engine
       -> Goal Calculation Engine
       -> Feasibility Engine
       -> Score Engine
       -> Insurance Engine
       -> Investment Allocation Engine
       -> Recommendation Engine
  -> Structured BoltPe Response
  -> Database
  -> PDF / S3 / Email
```

5. Functional Overview

Module	Responsibility
Customer Profile	Personal details, DOB-derived age, city, family state and dependents.
Cashflow	Income, expenses, savings, surplus and ratios.
Goal Engine	Future corpus, lump sum, monthly SIP/saving and timeline calculations.
Feasibility	Checks whether current savings can fund the selected timeline and derives achievable timeline.
Score	Produces financial score and status using BoltPe-owned weighted rules.
Insurance	Calculates recommended cover, existing cover, protection gap and gap status.
Investment	Allocates investable surplus by risk appetite and planning horizon.
Recommendations	Generates deterministic action points from calculated

	conditions.
Report	Produces normalized JSON and BoltPe PDF using the same calculation object.
Persistence	Stores request, assumptions, calculations, report state and audit fields.

6. User Journey

1. Customer enters personal details.
2. Customer selects one or more financial goals.
3. System captures family information only where relevant.
4. Customer enters monthly income, expenses and savings.
5. Customer provides risk appetite and investment experience.
6. Customer enters existing insurance cover information.
7. Customer enters or accepts goal assumptions such as inflation and expected return.
8. Backend validates and normalizes all values.
9. BoltPe calculation engine calculates each selected goal and the overall financial position.
10. System generates score, insurance gap, asset allocation, feasibility and recommendations.
11. Structured JSON is stored and returned.
12. PDF generation, storage and email may run synchronously or asynchronously depending on endpoint design.

7. Input Data Specification

7.1 Customer and Family Inputs

Field	Type	Required	Rule / Purpose
full_name	string	Yes	Customer name used in report and PDF.
contact	string	Yes	Validated mobile/contact number.
email	string	Conditional	Required when report email delivery is requested.
dob	date	Yes	Age is derived by backend; future DOB not allowed.
city	string	Yes	Used for profile and future location-based rules.
marital_status	enum	Conditional	single / married; may be skipped for instant-only plans if product flow allows.
spouse_age	integer	Conditional	Required for married customer where applicable.
spouse_income	decimal	No	Used in family-income analysis.
children_count	integer	No	Controls child-detail capture.
dependents	integer	No	Used in protection calculations.

7.2 Cashflow Inputs

Field	Type	Validation
monthly_income	decimal	> 0
monthly_expenses	decimal	>= 0 and normally <= monthly income

monthly_savings	decimal	>= 0

Derived metrics include:

- Disposable Income = Monthly Income - Monthly Expenses
- Savings Ratio = Monthly Savings / Monthly Income x 100
- Expense Ratio = Monthly Expenses / Monthly Income x 100
- Unallocated Cashflow = Monthly Income - Monthly Expenses - Monthly Savings

7.3 Risk and Insurance Inputs

Field Group	Values / Examples
risk_appetite	low / medium / high
investment_experience	beginner / intermediate / advanced
existing_life_term	boolean
existing_life_term_cover	amount
existing_health	boolean
existing_health_cover	amount
existing_critical	boolean
existing_critical_cover	amount
existing_personal	boolean
existing_personal_cover	amount

8. Goal Catalogue and Goal Rules

8.1 Instant Goals

Timeline: 6–12 months. Proposed BoltPe assumptions: Inflation 6% p.a.; Expected SIP Return 10% p.a.; Expected Lump Sum Return 10% p.a.

Examples: iPhone, PlayStation, Bike, TV, Fridge, Washing Machine, Vacation and Others.

BoltPe Instant Goals shall use the same inflation-and-return-based calculation framework as Short-Term and Long-Term Goals. Instant Goals support a 6–12 month timeline. The proposed configurable assumptions are 5% p.a. inflation, 6% p.a. expected SIP return and 6% p.a. expected lump-sum return. For month-based goals, Timeline in Years = Timeline Months / 12.

8.2 Short-Term Goals

Timeline: 3–5 years. Proposed BoltPe assumptions: Inflation 6% p.a.; Expected SIP Return 11% p.a.; Expected Lump Sum Return 10% p.a.

Examples: Self Development / Higher Education, Home Purchase / Home Upgrade, Car Purchase and Own Marriage Planning.

8.3 Long-Term Goals

Timeline: 20–25 years. Proposed BoltPe assumptions: Inflation 6% p.a.; Expected SIP Return 12% p.a.; Expected Lump Sum Return 12% p.a.

Examples: Retirement Planning and Wealth Creation.

8.4 Canonical Goal Model

```
{
  "key": "car_purchase",
  "label": "Car Purchase",
  "type": "short",
  "categoryTitle": "Short Term Goals",
  "timeline": "Within 3 years",
  "amount": 1000000,
  "years": 3,
  "months": 0,
  "customName": ""
}
```

8.5 Goal Timeline and Assumption Matrix

Goal Type	Timeline	Inflation	Expected SIP Return	Expected Lump Sum Return
Instant Goal	6–12 months	6% p.a.	10% p.a.	10% p.a.
Short-Term Goal	3–5 years	6% p.a.	11% p.a.	11% p.a.
Long-Term Goal	20–25 years	6% p.a.	12% p.a.	12% p.a.

9. Goal Calculation Logic

9.1 Future Corpus

For inflation-based goals:

```
Future Corpus = Current Goal Cost x (1 + Inflation Rate)^Years
```

Example aligned with the current response: a car goal of ₹10,00,000 over 3 years at 6% inflation produces approximately ₹11.91 lakh future corpus.

9.2 Lump Sum Required Today

```
Lump Sum Today = Future Corpus / (1 + Expected Lump Sum Return)^Years
```

For the sample 3-year car goal, the existing response shows approximately ₹8.95 lakh lump sum required today.

9.3 Monthly SIP / Saving Required

The engine shall calculate the periodic contribution required to reach the future corpus. The contribution timing convention (beginning-of-month or end-of-month), compounding frequency, return conversion method and rounding rules must be explicitly fixed in the calculation specification.

$$\text{Monthly SIP} = \frac{FV}{\left(\frac{(1+r)^N - 1}{r}\right) \times (1+r)}$$

Where:

- **Monthly SIP / Saving Required** = Required monthly investment
- **FV** = Future Corpus Needed (Inflation Adjusted Goal Amount)
- **r** = Monthly Rate of Return

$$r = \frac{\text{Annual SIP Return}}{12 \times 100}$$

- **N** = Total Number of Months

$$N = (\text{Years} \times 12) + \text{Additional Months}$$

Equivalent Formula

$$\text{Monthly SIP} = \frac{FV \times r}{((1+r)^N - 1) \times (1+r)}$$

10. Feasibility and Income Gap Analysis

For every goal the engine shall compare the required monthly contribution with available monthly savings. The result should preserve the current response fields.

Output Field	Description
Current Monthly Savings	Savings amount considered for feasibility.
Selected Timeline	Timeline chosen by customer.
Achievable Timeline With Current Savings	Timeline derived from available contribution.
Extra Time Required	Difference where current savings cannot meet the selected timeline.
Required Monthly Income To Keep Same Timeline	Income required under the approved affordability model.
Advisory Note	Human-readable conclusion based on feasibility result.

Parity Rule	The formulas behind achievable timeline and required monthly income are not fully exposed by the response. They must be reverse-engineered from multiple vendor examples and frozen into BoltPe-owned rules before exact-parity sign-off.

11. Insurance and Protection Analysis

The existing response contains separate rows for Life / Term Insurance, Health Insurance, Critical Illness Insurance and Personal Accident Insurance, along with recommended cover, existing cover, protection gap and gap status.

Insurance Type	Required Outputs
Life / Term	Recommended Cover, Existing Cover, Protection Gap, Gap Status, Logic
Health	Recommended Cover, Existing Cover, Protection Gap, Gap Status, Logic

Critical Illness	Recommended Cover, Existing Cover, Protection Gap, Gap Status, Logic
Personal Accident	Recommended Cover, Existing Cover, Protection Gap, Gap Status, Logic

Proposed configurable life-cover model:

```
Recommended Life Cover = (Annual Family Income x Income Multiple) + Liabilities + Major Family Goals - Eligible Existing Financial Assets
```

The exact multipliers and health/critical/personal-accident rules used for parity must be calibrated from test responses and approved by the product team.

12. Investment Allocation Engine

The sample response for medium risk appetite contains an allocation of 60% Equity, 30% Debt and 10% Gold / Alternatives. BoltPe should store such allocation policies in configuration rather than hardcode them in report code.

Risk Appetite	Illustrative Allocation
Low	Lower equity, higher debt/liquid exposure
Medium	60% Equity / 30% Debt / 10% Gold or approved equivalent
High	Higher equity exposure with horizon safeguards

Goal horizon must override or constrain generic risk allocation for very short-term goals.

12.1 Monthly Surplus Allocation Policy

- Emergency Fund: **20%**
- Goal Funding SIP: **50%**
- Long-Term Wealth Creation: **30%**

13. Recommendation and Priority Engine

Recommendations shall be deterministic and generated from calculated conditions.

Condition	Example Recommendation
Required SIP > Current Savings	Extend the timeline or increase monthly surplus/income.
Insurance Gap > 0	Review protection cover for the affected insurance category.
Emergency reserve below target	Build emergency fund separately from long-term investments.
Income increases	Use annual step-up SIP rather than keeping contribution unchanged.
Goal assumptions are old	Review goal values annually for inflation, income and family changes.

14. Retirement and Wealth Planning

Retirement and Wealth goals are treated as **Long-Term Goals**, but they may use **goal-specific assumptions** instead of the default Long-Term assumptions. The current request structure supports retirement inflation, pre-retirement return, post-retirement return, wealth inflation, wealth lump-sum return, and wealth SIP return.

BoltPe_Financial_Planner_FSD_v1.0.docx

Retirement Planning

- Retirement Planning shall use **retirement-specific assumptions** when values are provided in the request.
- The following request fields shall be used during the retirement corpus accumulation phase:
  - retirement_inflation
  - retirement_pre_return
- The retirement_post_return field is reserved for future retirement income and post-retirement cashflow calculations and is **not used** in the current corpus calculation model.
- If any retirement-specific assumption is **missing, empty, or zero**, the system shall use the default Long-Term assumptions:
  - Inflation: **6% p.a.**
  - Expected SIP Return: **12% p.a.**
  - Expected Lump Sum Return: **12% p.a.**

Wealth Creation

- Wealth Creation shall use **wealth-specific assumptions** when values are provided in the request.
- The following request fields shall be used:
  - wealth_inflation
  - wealth_sip_return
  - wealth_lumpsum_return
- If any wealth-specific assumption is **missing, empty, or zero**, the system shall use the default Long-Term assumptions:
  - Inflation: **6% p.a.**
  - Expected SIP Return: **12% p.a.**
  - Expected Lump Sum Return: **12% p.a.**

15. Output / Response Contract

The initial BoltPe-owned engine should preserve the normalized structure used by the current application to minimize frontend and PDF changes.

```
{
  "customer": {...},
  "score": {...},
```

```
  "summary": {...},
  "priority_roadmap": [...],
  "reports": [
    {
      "type": "...",
      "title": "...",
      "sub_title": "...",
      "summary": {...},
      "assumptions": {...},
      "table": [...],
      "feasibility": {...},
      "recommendations": [...]
    }
  ],
  "insurance_investment": {...},
  "common_recommendations": [...]
}
```

Vendor-specific company contact fields and AI/vendor branding should be removed or replaced with BoltPe-owned content.

16. Appendix - Sample Calculations

Calculation basis: Future Corpus is inflation-adjusted; Lump Sum Today is discounted at the expected annual lump-sum return; Monthly SIP uses the annuity-due formula with monthly rate = annual SIP return ÷ 12 and N = total months. Values are rounded to the nearest rupee.

16.1 Instant Goal

Input / Output	Value
Goal	iPhone
Present Goal Cost (PV)	₹90,000
Timeline	12 months = 1 year
Inflation Rate (i)	6% p.a.
Future Corpus Formula	FV = PV × (1 + i)^n
Future Corpus Calculation	₹90,000 × $(1 + 0.06)^1$ = ₹95,400
Future Corpus Needed (FV)	₹95,400
Expected Lump Sum Return (R)	10% p.a.
Lump Sum Formula	PV = FV ÷ (1 + R)^n
Lump Sum Calculation	₹95,400 ÷ $(1 + 0.10)^1$ = ₹86,727
Lump Sum Needed Today	₹86,727
Expected SIP Return	10% p.a.
Monthly Rate (r)	10% ÷ 12 = 0.8333% = 0.008333
Total Months (N)	12
Monthly SIP Convention	Beginning-of-month contribution (Annuity Due)
Monthly SIP Formula	SIP = FV / [((1+r)^N − 1)/r × (1+r)]
Monthly SIP Required	≈ ₹7,529 per month

16.2 Short-Term Goal

Input / Output	Value
Goal	Car Purchase
Present Goal Cost (PV)	₹10,00,000
Timeline	3 years = 36 months
Inflation Rate (i)	6% p.a.
Future Corpus Formula	FV = PV × (1 + i)^n
Future Corpus Calculation	₹10,00,000 × $(1 + 0.06)^3$ = ₹11,91,016
Future Corpus Needed (FV)	₹11,91,016 (Approx. ₹11.91 lakh)
Expected Lump Sum Return (R)	11% p.a.
Lump Sum Formula	PV_0 = FV ÷ (1 + R)^n
Lump Sum Calculation	₹11,91,016 ÷ $(1 + 0.11)^3$ = ₹8,70,861
Lump Sum Needed Today	₹8,71,550 (Approx. ₹8.72 lakh)
Expected SIP Return	11% p.a.
Monthly Rate (r)	11% ÷ 12 = 0.9167% = 0.009167
Total Months (N)	36
Monthly SIP Convention	Beginning-of-month contribution (Annuity Due)
Monthly SIP Formula	SIP = FV / [((1+r)^N − 1)/r × (1+r)]
Monthly SIP Required	≈ ₹27,820 per month

16.3 Long-Term Goal

Input / Output	Value
Goal	Retirement Planning
Present Target Amount (PV)	₹2,00,00,000 (₹2.00 Crore)
Timeline	25 years = 300 months
Inflation Rate (i)	6% p.a.
Future Corpus Formula	FV = PV × (1 + i)^n
Future Corpus Calculation	₹2,00,00,000 × (1 + 0.06)^25 = ₹8,58,37,414
Future Corpus Needed (FV)	₹8,58,37,414 (Approx. ₹8.58 Crore)
Expected Lump Sum Return (R)	12% p.a.
Lump Sum Formula	PV_0 = FV ÷ (1 + R)^n
Lump Sum Calculation	₹8,58,37,414 ÷ (1 + 0.12)^25 = ₹50,49,241
Lump Sum Needed Today	₹50,49,241 (Approx. ₹50.49 Lakh)
Expected SIP Return	12% p.a.
Monthly Rate (r)	12% ÷ 12 = 1% = 0.01
Total Months (N)	300
Monthly SIP Convention	Beginning-of-month contribution (Annuity Due)
Monthly SIP Formula	SIP = FV / [((1+r)^N − 1)/r × (1+r)]
Monthly SIP Required	₹45,234 per month